

[RCA] 泰康 Broker Load 报错 Broker list path exception

问题现象

- 用户在执行需要访问 HDFS 的任务时，前端看到失败原因为： Broker list path

```
exception
[tkamc@zcv120-vmware log]$ grep EasyTransfer202508183338e5115fc646789894fb0c09fc0e29 fe.log
2025-08-18 10:25:05,265 INFO (Load job scheduler|33) [DatabaseTransactionMgr.beginTransaction():313] begin transaction: txn_id: 359234413 with label EasyTransfer202508183338e5115fc646789894fb0c09fc0e29 from coordinator FE: 10.89.29.202, listner id: 215390552
2025-08-18 10:25:05,535 INFO (pending_load_task_scheduler_pool-0|11946) [DatabaseTransactionMgr.abortTransaction():1289] transaction:[TransactionState. txn_id: 359234413, label: EasyTransfer202508183338e5115fc646789894fb0c09fc0e29, db id: 2643631, table id list: 198921841, callback id: 215390552, coordinator: FE: 10.89.29.202, transaction status: ABORTED, error replicas num: 0, replica ids: , prepare time: 1755483905265, commit time: -1, finish time: 1755483905532, total cost: 267ms, reason: Broker list path exception. path=hdfs://easyops-cluster/user/dataworks/hive_db/datasec.db/app_spread_bond_ha/part-01852-2d119815-8599-47bc-9522-8737eca7553f-c000, broker=TNetworkAddress(hostname:10.89.24.121, port:8000) exception: java.net.SocketException: Connection reset] successfully rollback
2025-08-18 10:25:49,031 INFO (thrift-server-pool-78847612|230485625) [StmtExecutor.execute():605] execute Exception, sql: show load where label = "EasyTransfer202508183338e5115fc646789894fb0c09fc0e29", error: No database selected
2025-08-18 10:40:15,999 INFO (Load job scheduler|33) [DatabaseTransactionMgr.beginTransaction():313] begin transaction: txn_id: 359266481 with label EasyTransfer202508183338e5115fc646789894fb0c09fc0e29 from coordinator FE: 10.89.29.202, listner id: 215402985
2025-08-18 10:40:16,502 INFO (pending_load_task_scheduler_pool-1|919454) [DatabaseTransactionMgr.abortTransaction():1289] transaction:[TransactionState. txn_id: 359266481, label: EasyTransfer202508183338e5115fc646789894fb0c09fc0e29, db id: 2643631, table id list: 198921841, callback id: 215402985, coordinator: FE: 10.89.29.202, transaction status: ABORTED, error replicas num: 0, replica ids: , prepare time: 1755484815999, commit time: -1, finish time: 1755484816500, total cost: 501ms, reason: Broker list path exception. path=hdfs://easyops-cluster/user/dataworks/hive_db/datasec.db/app_spread_bond_ha/part-03176-2d119815-8599-47bc-9522-8737eca7553f-c000, broker=TNetworkAddress(hostname:10.89.24.121, port:8000) exception: java.net.SocketException: Connection reset] successfully rollback
[tkamc@zcv120-vmware log]$
```

代码块

```
1 2025-08-18 10:25:05,532 WARN (pending_load_task_scheduler_pool-0|11946)
  [LoadTask.exec():66] LOAD_JOB=215390552, error_msg={Failed to execute load
  task}
2 com.starrocks.common.UserException: Broker list path exception.
  path=hdfs://easyops-
  cluster/user/dataworks/hive_db/datasec.db/app_spread_bond_ha/part-01852-
  2d119815-8599-47bc-9522-
    8737eca7553f-c000,
  broker=TNetworkAddress(hostname:10.89.24.121, port:8000) exception:
  java.net.SocketException: Connection reset
3   at com.starrocks.common.util.BrokerUtil.parseFile(BrokerUtil.java:111) ~
  [starrocks-fe.jar:?]
4   at
  com.starrocks.load.loadv2.BrokerLoadPendingTask.getAllFileStatus(BrokerLoadPend
  ingTask.java:82) ~[starrocks-fe.jar:?]
5   at
  com.starrocks.load.loadv2.BrokerLoadPendingTask.executeTask(BrokerLoadPendingTa
  sk.java:62) ~[starrocks-fe.jar:?]
6   at com.starrocks.load.loadv2.LoadTask.exec(LoadTask.java:60) ~[starrocks-
  fe.jar:?]
7   at com.starrocks.task.PriorityLeaderTask.run(PriorityLeaderTask.java:25) ~
  [starrocks-fe.jar:?]
8   at
  java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:511) ~
  [?:1.8.0_322]
```

```

9      at java.util.concurrent.FutureTask.run(FutureTask.java:266) ~[?:1.8.0_322]
10     at
java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1149)
~[?:1.8.0_322]
11     at
java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:624)
~[?:1.8.0_322]
12     at java.lang.Thread.run(Thread.java:750) ~[?:1.8.0_322]

```

更早时间还有其它 `Read timed out` 报错

代码块

```

1    2025-08-17 02:37:23,612 WARN (pending_load_task_scheduler_pool-0|11946)
[LoadTask.exec():66] LOAD_JOB=215085294, error_msg={Failed to execute load
task}
2    com.starrocks.common.UserException: Broker list path exception.
path=hdfs://easyops-cluster/user/fidm/easytransfer-806fd503-b44a-465e-a092-
80b6f713a114/part-00000-58c0ebdb-1d9c-4b0d-      832c-9d47a34059c8-
c000.gz.parquet, broker=TNetworkAddress(hostname:10.89.24.121, port:8000)
exception: java.net.SocketTimeoutException: Read timed out
3      at com.starrocks.common.util.BrokerUtil.parseFile(BrokerUtil.java:111) ~
[starrocks-fe.jar:?]
4      at
com.starrocks.load.loadv2.BrokerLoadPendingTask.getAllFileStatus(BrokerLoadPend
ingTask.java:82) ~[starrocks-fe.jar:?]
5      at
com.starrocks.load.loadv2.BrokerLoadPendingTask.executeTask(BrokerLoadPendingTa
sk.java:62) ~[starrocks-fe.jar:?]
6      at com.starrocks.load.loadv2.LoadTask.exec(LoadTask.java:60) ~[starrocks-
fe.jar:?]
7      at com.starrocks.task.PriorityLeaderTask.run(PriorityLeaderTask.java:25) ~
[starrocks-fe.jar:?]
8      at
java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:511) ~
[?:1.8.0_322]
9      at java.util.concurrent.FutureTask.run(FutureTask.java:266) ~[?:1.8.0_322]
10     at
java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1149)
~[?:1.8.0_322]
11     at
java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:624)
~[?:1.8.0_322]
12     at java.lang.Thread.run(Thread.java:750) ~[?:1.8.0_322]

```

- 同期 jstack 中可见多条线程长期卡住:

- 业务线程在 `listPath` 调用链上阻塞于 `ReentrantLock.lock()` (`FileSystemManager.getDistributedFileSystem`)，这是导致 Broker Load 报错 list path 失败的直接原因

代码块

```
1  "TThreadPoolServer WorkerProcess-%d" #2012419 daemon prio=5 os_prio=0
   tid=0x000007f6098029800 nid=0x1503c waiting on condition [0x000007f61aa0e7000]
2      java.lang.Thread.State: WAITING (parking)
3          at sun.misc.Unsafe.park(Native Method)
4              - parking to wait for <0x00000000c0f190c0> (a
   java.util.concurrent.locks.ReentrantLock$NonfairSync)
5          at java.util.concurrent.locks.LockSupport.park(LockSupport.java:175)
6          at
   java.util.concurrent.locks.AbstractQueuedSynchronizer.parkAndCheckInterrupt(
   AbstractQueuedSynchronizer.java:836)
7          at
   java.util.concurrent.locks.AbstractQueuedSynchronizer.acquireQueued(Abstract
   QueuedSynchronizer.java:870)
8          at
   java.util.concurrent.locks.AbstractQueuedSynchronizer.acquire(AbstractQueued
   Synchronizer.java:1199)
9          at
   java.util.concurrent.locks.ReentrantLock$NonfairSync.lock(ReentrantLock.java
   :209)
10         at
   java.util.concurrent.locks.ReentrantLock.lock(ReentrantLock.java:285)
11         at
   com.starrocks.broker.hdfs.FileSystemManager.getDistributedFileSystem(FileSys
   temManager.java:303)
12         at
   com.starrocks.broker.hdfs.FileSystemManager.getFileSystem(FileSystemManager.
   java:199)
13         at
   com.starrocks.broker.hdfs.FileSystemManager.listPath(FileSystemManager.java:
   796)
14         at
   com.starrocks.broker.hdfs.HDFSBrokerServiceImpl.listPath(HDFSBrokerServiceIm
   pl.java:74)
15         at
   com.starrocks.thrift.TFileBrokerService$Processor$listPath.getResult(TFileBr
   okerService.java:815)
16         at
   com.starrocks.thrift.TFileBrokerService$Processor$listPath.getResult(TFileBr
```

```

    okerService.java:795)
17      at org.apache.thrift.ProcessFunction.process(ProcessFunction.java:38)
18      at org.apache.thrift.TBaseProcessor.process(TBaseProcessor.java:38)
19      at
    org.apache.thrift.server.TThreadPoolServer$WorkerProcess.run(TThreadPoolServ
    er.java:248)
20      at
    java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:11
    49)
21      at
    java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:6
    24)
22      at java.lang.Thread.run(Thread.java:750)

```

- File system 回收线程在 Hadoop RPC 停机路径里 WAITING
(`org.apache.hadoop.ipc.Client.stop()`)。

代码块

```

1  "pool-1-thread-2" #25 prio=5 os_prio=0 tid=0x00007f627cba7000 nid=0xe358
    in Object.wait() [0x00007f61b8190000]
2      java.lang.Thread.State: WAITING (on object monitor)
3      at java.lang.Object.wait(Native Method)
4      at java.lang.Object.wait(Object.java:502)
5      at org.apache.hadoop.ipc.Client.stop(Client.java:1375)
6      - locked <0x00000000c0f20b30> (a java.lang.Object)
7      at org.apache.hadoop.ipc.ClientCache.stopClient(ClientCache.java:118)
8      at
    org.apache.hadoop.ipc.ProtobufRpcEngine2$Invoker.close(ProtobufRpcEngine2.ja
    va:339)
9      at org.apache.hadoop.ipc.RPC.stopProxy(RPC.java:804)
10     at
    org.apache.hadoop.hdfs.protocolPB.ClientNamenodeProtocolTranslatorPB.close(C
    llientNamenodeProtocolTranslatorPB.java:321)
11     at
    org.apache.hadoop.hdfs.server.namenode.ha.ConfiguredFailoverProxyProvider.cl
    ose(ConfiguredFailoverProxyProvider.java:81)
12     - locked <0x00000000c0f20dd8> (a
    org.apache.hadoop.hdfs.server.namenode.ha.ConfiguredFailoverProxyProvider)
13     at
    org.apache.hadoop.io.retry.RetryInvocationHandler$ProxyDescriptor.close(Retr
    yInvocationHandler.java:235)
14     at
    org.apache.hadoop.io.retry.RetryInvocationHandler.close(RetryInvocationHandl
    er.java:455)
15     at org.apache.hadoop.ipc.RPC.stopProxy(RPC.java:804)

```

```

16         at
org.apache.hadoop.hdfs.DFSCClient.closeConnectionToNamenode(DFSCClient.java:62
1)
17         at org.apache.hadoop.hdfs.DFSCClient.close(DFSCClient.java:667)
18         - locked <0x00000000c0f20b90> (a org.apache.hadoop.hdfs.DFSCClient)
19         at
org.apache.hadoop.hdfs.DistributedFileSystem.close(DistributedFileSystem.jav
a:1534)
20         at
com.starrocks.broker.hdfs.BrokerFileSystem.closeFileSystem(BrokerFileSystem.
java:62)
21         at
com.starrocks.broker.hdfs.FileSystemManager$FileSystemExpirationChecker.run(
FileSystemManager.java:1055)
22         at
java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:511)
23         at java.util.concurrent.FutureTask.run(FutureTask.java:266)
24         at
java.util.concurrent.ScheduledThreadPoolExecutor$ScheduledFutureTask.access$
201(ScheduledThreadPoolExecutor.java:180)
25         at
java.util.concurrent.ScheduledThreadPoolExecutor$ScheduledFutureTask.run(Sch
eduledThreadPoolExecutor.java:293)
26         at
java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:11
49)
27         at
java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:6
24)
28         at java.lang.Thread.run(Thread.java:750)

```

- `openReader` 线程在发起 HDFS RPC（如 `getBlockLocations`）时卡在 `SynchronousQueue.offer()`（`Client$Connection.sendRpcRequest`）

代码块

```

1  "TThreadPoolServer WorkerProcess-%d" #2012398 daemon prio=5 os_prio=0
tid=0x000007f6098024800 nid=0x4b71c waiting on condition [0x000007f61aaef1000]
2      java.lang.Thread.State: TIMED_WAITING (parking)
3          at sun.misc.Unsafe.park(Native Method)
4          - parking to wait for <0x00000000c48c4950> (a
java.util.concurrent.SynchronousQueue$TransferQueue)
5          at
java.util.concurrent.locks.LockSupport.parkNanos(LockSupport.java:215)
6          at
java.util.concurrent.SynchronousQueue$TransferQueue.awaitFulfill(Synchronous

```

```
Queue.java:764)
7      at
  java.util.concurrent.SynchronousQueue$TransferQueue.transfer(SynchronousQueue
e.java:695)
8      at
  java.util.concurrent.SynchronousQueue.offer(SynchronousQueue.java:895)
9      at
  org.apache.hadoop.ipc.Client$Connection.sendRpcRequest(Client.java:1171)
10     at org.apache.hadoop.ipc.Client.call(Client.java:1463)
11     at org.apache.hadoop.ipc.Client.call(Client.java:1410)
12     at
  org.apache.hadoop.ipc.ProtobufRpcEngine2$Invoker.invoke(ProtobufRpcEngine2.j
ava:258)
13     at
  org.apache.hadoop.ipc.ProtobufRpcEngine2$Invoker.invoke(ProtobufRpcEngine2.j
ava:139)
14     at com.sun.proxy.$Proxy14.getBlockLocations(Unknown Source)
15     at
  org.apache.hadoop.hdfs.protocolPB.ClientNamenodeProtocolTranslatorPB.getBloc
kLocations(ClientNamenodeProtocolTranslatorPB.java:334)
16     at sun.reflect.GeneratedMethodAccessor19.invoke(Unknown Source)
17     at
  sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl
.java:43)
18     at java.lang.reflect.Method.invoke(Method.java:498)
19     at
  org.apache.hadoop.io.retry.RetryInvocationHandler.invokeMethod(RetryInvocati
onHandler.java:433)
20     at
  org.apache.hadoop.io.retry.RetryInvocationHandler$Call.invokeMethod(RetryInv
ocationHandler.java:166)
21     at
  org.apache.hadoop.io.retry.RetryInvocationHandler$Call.invoke(RetryInvocatio
nHandler.java:158)
22     at
  org.apache.hadoop.io.retry.RetryInvocationHandler$Call.invokeOnce(RetryInvoc
ationHandler.java:96)
23     - locked <0x00000000c494c350> (a
  org.apache.hadoop.io.retry.RetryInvocationHandler$Call)
24     at
  org.apache.hadoop.io.retry.RetryInvocationHandler.invoke(RetryInvocationHand
ler.java:362)
25     at com.sun.proxy.$Proxy15.getBlockLocations(Unknown Source)
26     at
  org.apache.hadoop.hdfs.DFSClient.callGetBlockLocations(DFSClient.java:900)
27     at
  org.apache.hadoop.hdfs.DFSClient.getLocatedBlocks(DFSClient.java:889)
```

```

28         at
org.apache.hadoop.hdfs.DFSClient.getLocatedBlocks(DFSClient.java:878)
29         at org.apache.hadoop.hdfs.DFSClient.open(DFSClient.java:1046)
30         at
org.apache.hadoop.hdfs.DistributedFileSystem$4.doCall(DistributedFileSystem.
java:343)
31         at
org.apache.hadoop.hdfs.DistributedFileSystem$4.doCall(DistributedFileSystem.
java:339)
32         at
org.apache.hadoop.fs.FileSystemLinkResolver.resolve(FileSystemLinkResolver.j
ava:81)
33         at
org.apache.hadoop.hdfs.DistributedFileSystem.open(DistributedFileSystem.java
:356)
34         at
com.starrocks.broker.hdfs.FileSystemManager.openReader(FileSystemManager.jav
a:891)
35         at
com.starrocks.broker.hdfs.HDFSBrokerServiceImpl.openReader(HDFSBrokerService
Impl.java:138)
36         at
com.starrocks.thrift.TFileBrokerService$Processor$openReader.getResult(TFile
BrokerService.java:915)
37         at
com.starrocks.thrift.TFileBrokerService$Processor$openReader.getResult(TFile
BrokerService.java:895)
38         at org.apache.thrift.ProcessFunction.process(ProcessFunction.java:38)
39         at org.apache.thrift.TBaseProcessor.process(TBaseProcessor.java:38)
40         at
org.apache.thrift.server.TThreadPoolServer$WorkerProcess.run(TThreadPoolServ
er.java:248)
41         at
java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:11
49)
42         at
java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:6
24)
43         at java.lang.Thread.run(Thread.java:750)

```

原因分析

- 触发条件

- Broker 内有一个后台过期回收任务 `FileSystemExpirationChecker`，定期回收过期的 `BrokerFileSystem`。
- 回收时，它持有 `BrokerFileSystem` 的互斥锁，直接调用
- `DistributedFileSystem.close()` 会关闭 `DFSClient` → 停掉底层 RPC 代理（`RPC.stopProxy`）→ 通过 `ClientCache.stopClient` 触发 `Client.stop()`。
- Hadoop 3.3.6 的 `Client.stop()` 逻辑是：中断每个连接的发送/连接线程，然后在 `emptyCondition.wait()` 等待“所有连接关闭”。此时连接的发送线程已被中断，不再从 `SynchronousQueue` 消费新请求。
- 并发表现
 - 当回收线程拿着 `BrokerFileSystem` 的锁执行关闭时，`listPath` 线程需要先拿到同一把锁进入 `getFileSystem()`，因此被阻塞在 `ReentrantLock.lock()` 上，形成在 jstack 中看到的 WAITING (parking)。
 - 与此同时，`openReader` 线程还在向同一 RPC 连接发起新调用（如 `getBlockLocations`），其发送路径位于 `Client$Connection.sendRpcRequest()`，需要把消息放进一个 `SynchronousQueue` 交给连接的发送线程。但发送线程已被 `Client.stop()` 中断，不再消费，导致调用线程在 `offer()` 上进行 1s 重试式等待，呈现 TIMED_WAITING。
 - 连接未能完成“关闭”收尾（因为仍有未完成调用/锁竞争），`Client.stop()` 就会一直在 `emptyCondition.wait()` 里等，后台回收线程占着 `BrokerFileSystem` 的锁不放，前台业务线程又拿不到锁，形成环状僵持。
- 代码证据（关键位置）
 - Broker 侧持锁关闭 HDFS：

代码块

```

1  class FileSystemExpirationChecker implements Runnable {
2      public void run() {
3          for (BrokerFileSystem fs : cachedFileSystem.values()) {
4              if (fs.isExpired()) {
5                  fs.getLock().lock();
6                  try {
7                      fs.closeFileSystem();        // 持锁调用 close()
8                  } finally {
9                      cachedFileSystem.remove(fs.getIdentity());
10                     fs.getLock().unlock();
11                 }
12             }
13         }
14     }
15 }

```



```

14      ...
15    }
16  }

```

代码块

```

1  public void closeFileSystem() {
2      lock.lock();
3      try {
4          if (dfsFileSystem != null) {
5              dfsFileSystem.close();           // 进入 Hadoop 关闭链路
6          }
7      } finally {
8          lock.unlock();
9      }
10 }

```

- `listPath` 在 `getFileSystem` 上等待这把锁：

代码块

```

1  fileSystem.getLock().lock(); // 这里与回收线程争用
2  try {
3      if (!cachedFileSystem.containsKey(fileSystemIdentity)) { ... }
4      ...

```

- Hadoop RPC 发送路径卡在 `SynchronousQueue.offer()`：

代码块

```

1  while (!shouldCloseConnection.get()) {
2      if (rpcRequestQueue.offer(Pair.of(call, buf), 1, TimeUnit.SECONDS)) {
3          break;
4      }
5  }

```

- `Client.stop()` 中断发送线程并等待全部连接关闭：

代码块

```

1  for (Connection conn : connections.values()) {
2      conn.interrupt();
3      conn.rpcRequestThread.interrupt();
4      conn.interruptConnectingThread();

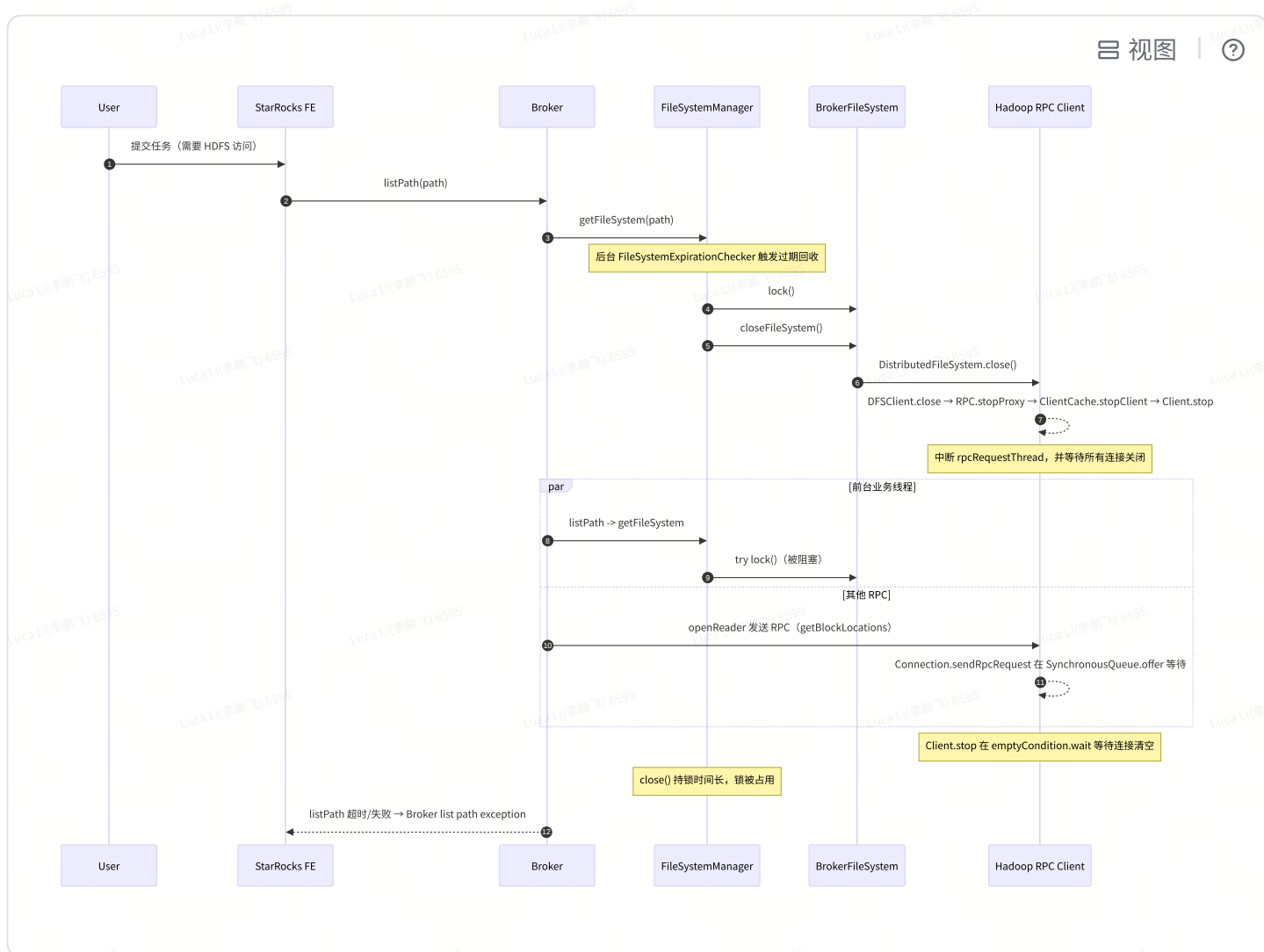
```

```

5  }
6  synchronized (emptyCondition) {
7      while (!connections.isEmpty()) {
8          emptyCondition.wait();
9      }
10 }

```

• 触发流程时序（示意）



• 结论

- 这三类卡栈互为因果：后台回收持锁关闭 HDFS → 触发 Hadoop RPC 停机等待；
openReader 新的 RPC 因发送线程已停而阻塞；listPath 线程在获取相同 BrokerFileSystem 的锁时被阻塞。最终在超时窗口内返回“Broker list path exception”。

修复方案

TBD